

View Reviews

Paper ID

181

Paper Title

VEduTutorBench: a High-Quality Vietnamese Secondary-school Informatics AI Tutor Benchmark from Teacher-Authored Dialogues

Track Name

Special Session - AI4S 2026

Reviewer #1

Questions**1. Comments to Authors**

VEduTutorBench provides a highly rigorous, curriculum-grounded benchmark for Vietnamese Informatics AI tutoring using 1,400 validated samples. While mapping Allison and Tharby's principles to tutoring capabilities is theoretically robust, the sample scarcity in "Challenge" (8 samples) and "Practice" (27 samples) limits your principle-level claims. Moreover, conducting a pilot human evaluation with expert teachers will be vital to calibrate the LLM judges and rule out potential same-family biases.

In addition, I have questions, involving the pipeline for VEduTutorBench (Figure 1): What is the relationship between Phase 1 (Data Audit) and Phase 2 (Measurement Foundation)? Are they can be performed in parallel?

3. Overall Score

Weak Accept

Reviewer #2

Questions**1. Comments to Authors**

This paper presents VEduTutorBench, a benchmark for evaluating LLM-based tutors in Vietnamese secondary-school Informatics for Grades 6–9. I consider the target setting relevant and relatively underexplored. The main strengths are the systematic and traceable construction pipeline, preservation of native multi-turn context, Vietnamese curriculum grounding, and the use of pedagogically motivated evaluation criteria. The resulting benchmark contains 1,400 samples from 665 dialogue families.

The experimental section is also useful in that it reports results from two LLM judges, family-level aggregation, confidence intervals, and cross-judge agreement. The authors are commendably explicit about several limitations of the current benchmark. Overall, I view this as a useful contribution to Vietnamese educational AI. I have four main suggestions that would further strengthen the paper.

1. Clarify the scope of pedagogical strategy selection.

The paper presents pedagogical strategy selection as an important AI-tutor capability. However, during response generation, the evaluated tutor is explicitly given the required pedagogical principles. Therefore, the current setting primarily measures the model's ability to realize a specified pedagogical strategy rather than fully evaluating autonomous strategy selection from the learner state. I suggest making this distinction explicit in the benchmark description and Discussion. An autonomous setting in which the model must infer the appropriate pedagogical move without being given the principle would be a valuable future extension.

2. Human validation would substantially strengthen the benchmark.

The current evaluation relies on two LLM judges. This provides useful evidence of cross-judge stability, but does not fully replace expert pedagogical evaluation. The paper appropriately acknowledges that LLM–LLM agreement does not establish correctness relative to an independent human reference. This is particularly relevant because one judge belongs to the Gemini family while two evaluated tutor configurations are also Gemini-based, and the reported evaluation of Llama differs noticeably between the Gemini and GPT judges. I encourage the authors to add, or clearly plan, a stratified human-evaluation subset reporting human–LLM agreement and analysis of judge disagreements.

3. The pedagogical-principle distribution is highly imbalanced.

Explanation, Feedback, and Questioning are relatively well represented, whereas Modelling contains 93 samples, Practice 27, and Challenge only 8. Consequently, principle-level conclusions for Challenge and Practice should be interpreted cautiously. I recommend explicitly treating these results as preliminary and prioritizing additional samples for the underrepresented principles in future versions of the benchmark.

4. Reproducibility and release details should be made more explicit.

As this is a benchmark paper, the final version should clearly specify which artifacts will be released, including benchmark samples, fixed splits, tutor and judge prompts, principle-scoring prompts, evaluation scripts, model versions, and decoding settings. If some curriculum-derived SGK/SGV materials cannot be redistributed because of copyright restrictions, the authors should explain how the benchmark and evaluation pipeline can nevertheless be reproduced.

A few smaller issues, such as the terminology used for “Accuracy”, the interpretation of “teacher-authored dialogues”, and the bootstrap clustering unit, could also be clarified in the final version.

Overall, I find VEduTutorBench to be a promising and practically useful benchmark,

especially for Vietnamese educational AI. The remaining limitations mainly concern the extent of validation and the interpretation of some benchmark claims rather than the core contribution itself. I therefore lean toward acceptance and encourage the authors to clarify the above issues in the revised version.

3. Overall Score

Weak Accept

Reviewer #3

Questions

1. Comments to Authors

The paper presents a Vietnamese Grades 6-9 Informatics AI-tutor benchmark built through a traceable pipeline from 1,050 teacher-authored dialogues. The pipeline audits the dialogues, establishes six pedagogical principles and six tutor capabilities, and converts eligible tutor turns into 2,028 context-conditioned candidates from 665 retained dialogue families. The experiment shows that framework preserves native multi-turn roles and compares tutor generated and teacher-authored responses using four common criteria, three criteria per required principle, and criterion-level Win/Tie/Lose judgments with three tutor configurations two Gemini-based tutors and Llama 4 Maverick-whose responses are independently assessed by two LLM judges.

The results in the table III-V need to be more detailed explanation and typos need to be corrected.

3. Overall Score

Weak Accept